

Study Report: NIST Cybersecurity Framework - Cybersecurity Analyst

Written by Gagan Pasupuleti

Family: Cybersecurity Analyst
Level: Intermediate
Chapters: 7
Source: Reference: NIST Cybersecurity Framework - NIST

Official sources and free libraries

- <https://www.nist.gov/cyberframework>

Summary

Study report for *NIST Cybersecurity Framework* by NIST (Intermediate level) mapped to the Cybersecurity Analyst role. Identify, Protect, Detect, Respond, Recover - free official guide.

Chapters

Track: Book-Intro

Chapter 1: About NIST Cybersecurity Framework [Intermediate]

Chapter focus: About NIST Cybersecurity Framework** *(Level: Intermediate)

This study report summarizes *NIST Cybersecurity Framework* by NIST for the Cybersecurity Analyst role. The resource is rated Intermediate level. Identify, Protect, Detect, Respond, Recover - free official guide. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# NIST Cybersecurity Framework
# Author: NIST
# Role: Cybersecurity Analyst
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on NIST Cybersecurity Framework.

When to use it

When learning Cybersecurity Analyst skills at Intermediate level.

Where to use it

Cybersecurity Analyst training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role** *(Level: Intermediate)

Cybersecurity Analyst professionals use ideas from NIST Cybersecurity Framework to solve real workplace problems. Identify, Protect, Detect, Respond, Recover - free official guide. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Cybersecurity Analyst

Book focus: Identify, Protect, Detect, Respond, Recover - free official guide.

Recommended level: Intermediate

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Cybersecurity Analyst bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered** *(Level: Intermediate)

The main topics in NIST Cybersecurity Framework include practical concepts described as: Identify, Protect, Detect, Respond, Recover - free official guide. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Cybersecurity Analyst jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: NIST Cybersecurity Framework [Intermediate]

Chapter focus: NIST Cybersecurity Framework** *(Level: Intermediate)

NIST CSF functions: Identify (asset management), Protect (access control), Detect (anomalies), Respond (incidents), Recover (backups). Create Current Profile vs Target Profile gap reports for leadership funding asks.

Code Reference:

```
functions = ['Identify', 'Protect', 'Detect', 'Respond', 'Recover']
```

What it shows: CSF functions organize control maturity assessments.

Topic guide

What it actually is

NIST CSF provides a common risk management vocabulary.

When to use it

Enterprise security program design and board reporting.

Where to use it

US critical infrastructure and widely adopted globally.

Real use example

Gap analysis shows Detect immature-funds SIEM and SOC headcount.

Chapter 5: Applied: MITRE ATT&CK for Detection [Intermediate]

Chapter focus: MITRE ATT&CK for Detection *(Level: Intermediate)**

ATT&CK catalogs adversary tactics (Initial Access, Execution, Persistence?) and techniques (T1059 PowerShell). Map detections and purple team tests to technique IDs. Navigator layers visualize coverage gaps.

Code Reference:

```
technique = {'id': 'T1110.001', 'name': 'Password Guessing', 'tactic': 'Credential Access'}
```

What it shows: Technique metadata enriches SIEM alert descriptions.

Topic guide

What it actually is

MITRE ATT&CK standardizes threat behavior knowledge.

When to use it

Detection engineering, threat intel, and purple teaming.

Where to use it

SOCs building coverage dashboards for leadership.

Real use example

New rule detects T1110.001 brute force-closes Navigator gap for Credential Access.

Chapter 6: Applied: SIEM Architecture and Log Sources [Intermediate]

Chapter focus: SIEM Architecture and Log Sources *(Level: Intermediate)**

SIEM aggregates logs: AD, EDR, firewall, cloud trail, app logs. Normalization, retention, and correlation rules produce alerts. Tune noisy rules-alert fatigue causes missed true positives. Store raw logs immutably for forensics.

Code Reference:

```
# Sentinel table: SecurityEvent | where EventID == 4625 | summarize count() by Account
```

What it shows: Failed logon aggregation detects brute-force patterns.

Topic guide

What it actually is

SIEM is the central nervous system for security monitoring.

When to use it

24/7 SOC operations and compliance log retention.

Where to use it

Microsoft Sentinel, Splunk Enterprise Security, Elastic.

Real use example

Spike in 4625 events across VPN-correlation opens incident within minutes.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice** *(Level: Intermediate)

Finish NIST Cybersecurity Framework with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Cybersecurity Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.